

A rig is an input

311 rigs; 34–102 joints

1 Rest pose

Rest-pose skeleton
image placeholder

KTJD-17

17 channels / joint
per-cell standardise

1 rest
 x_{demo}
clean

x_t^{tgt}
240 frames
target only

$x_t = (1 - t)x_0 + tx_1; x_0 \sim \mathcal{N}(0, I).$
Joint semantics added at every frame.
Rest frame: never noised or dropped.

2 Joint descriptions

“the left claw on the
fifth digit of the
front foot”

LLM2Vec
frozen

4096

Linear
4096 $\rightarrow D$

Linear
17 $\rightarrow D$
+ pos.
+ target mask

[1 | 240] frames

3 Tree

Kinematic tree
+ rest offsets

8-d
Struct. MLP
zero-init.
output

Tree attention bias
– $\min(g_{ij}, 8)$
+ learned up/down
LCA tables (per head)

AdaLN
Temporal attn.
frames / joint

AdaLN
Spatial attn.
joints / frame

AdaLN
MLP
4D

14 DiT blocks, shared across rigs

$D = 896$; 14 heads; 303M parameters
Residual sublayers; Q/K RMSNorm

Caption c

“the juvenile elephant starts to walk and turns left”

LLM2Vec
pooled, frozen

4096

LayerNorm
+ MLP

Flow time t
embedding + MLP

$\oplus$
 c : shift / scale / gate

$T \times J \times 17$

LN + **AdaLN**
shift / scale
Linear $D \rightarrow 17$

$\hat{x}_1$
clean

20 Euler steps: x
caption CFG $s = 2$

undo statistics
rotations $\rightarrow$ rig FK
 $\rightarrow$ skin

Skinned character
image placeholder

$x \leftarrow x + (\hat{x}_1 - x) / (20 - i); \text{targets only; invalid cells} \rightarrow 0$

Target-only FM: $\mathcal{L} = \sum_g \gamma_g \sqrt{N_g / N} \bar{\ell}_g$ 9 groups; γ_g calibrated from energies.

Unseen rig: same inputs + per-cell statistics; same path, zero training.

Optional rank-64 LoRA from ~ 10 clips; folded into exported weights.